

accrued principal of \$1,026. A share will provide real income of $1026 \times 1.0075 = \$1,033.70$ in 2045. We require $39603.96 / 1033.70 = 38.31$ shares of the 2045 TIPS. The ask price is \$95.13, so the price we must pay today is $95.13 \times 1026 / 100 = \975.98 . The total cost for these shares is \$37,393.

This process continues, working backward to the present. In 2017, the ladder provides \$17,734 of real income as interest payments for bonds maturing in later years. This means that we only need to set aside an additional \$22,266 for the \$40,000 of spending in the first year.

This is the logic behind constructing the TIPS ladder. Note that to deal with the problem of missing maturities, we bought three years of spending with the 2029 TIPS and eight years of spending with the 2032 TIPS.

It is impossible for a TIPS ladder to have a cumulative cost column that correctly shows the costs for building shorter TIPS ladders in the same manner as we had for Treasury STRIPS. With the thirty-year ladder shown, part of the income available at earlier dates comes from coupon payments from longer-dated bonds that would not be part of a shorter ladder. Because the ladder is constructed beginning at its end, we would have to start from scratch if we chose a different end date. This was not the case for STRIPS, since there was no interest income to complicate the calculations for earlier years in the ladder.

The total cost for this TIPS ladder with wholesale pricing is \$1.069 million. For the \$40,000 inflation-adjusted spending, this represents an initial withdrawal rate of 3.74 percent. The real internal rate of return on the cash flows as they are spent is 0.76 percent. A thirty-year TIPS ladder is as close as we can get to a real-world “safe withdrawal rate” for thirty years of inflation-adjusted spending. At current interest rates, 3.74 percent is the number. A thirty-year TIPS ladder represents the “risk-free” way to support thirty years of real income. Spending more from an investment portfolio is based on a hope that a higher return can justify and sustain a higher spending rate. This is risky. It is important to note that with this bond ladder, nothing will be left at the end of the thirtieth year. The calculation is “risk free” for thirty years, but the possibility of living beyond thirty years must be considered. A TIPS ladder does not hedge longevity risk.